

Certified FCI-Free MPS Response for Analytic State-Averaged DMRG-CASSCF Gradients and Nonadiabatic Couplings

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Abstract

Analytic nuclear gradients and nonadiabatic coupling vectors are central quantities for conical-intersection optimization and nonadiabatic molecular dynamics with multireference wave functions. State-averaged complete-active-space self-consistent-field (SA-CASSCF) theory is robust near electronic degeneracies, but the conventional full-configuration-interaction (FCI) active-space representation limits the active spaces for which derivative quantities can be evaluated and validated. Here we report a certified, FCI-free matrix-product-state (MPS) response implementation for analytic SA-DMRG-CASSCF gradients and nonadiabatic couplings in a PySCF/pyblock2 workflow. The implementation combines three ingredients: an MPS-Krylov response backend that never stores the active-space response vector as a dense determinant array, an MPS-native cross-geometry overlap protocol for finite-difference NAC validation without determinant expansion, and an a posteriori response certificate based on the true residual of the original coupled response equation and root-projector leakage. The method is validated in two complementary regimes. Where FCI is available, public benchmark

systems verify energies, density matrices, gradients, derivative couplings, root assignment, and MPS response algebra against exact active-space references. Where FCI is impossible, internal consistency is assessed by finite differences of the same SA-DMRG-CASSCF energy surface and by determinant-free cross-geometry MPS overlaps. The largest planned beyond-FCI benchmark is an all-trans polyene CAS(20,20) π space with $\binom{20}{10}^2 = 3.41 \times 10^{10}$ active-space determinants, for which no dense FCI vector or FCI-overlap root tracker is used. A sweep-localized Schur-complement response route and a certified auto-response driver provide independent response solves and same-geometry warm starts; all accepted results are gated by the same true-residual certificate. The SHARC-compatible interface is retained as an output data contract for energies, dipoles, gradients, NACs, and phases, but the present work is not claimed to make long-time CAS(20,20) surface-hopping dynamics routine. Instead, it establishes and validates the FCI-free derivative machinery required for large-active-space nonadiabatic workflows.

1 Introduction

Nonadiabatic molecular processes require more electronic-structure information than adiabatic energies alone. Surface-hopping and related mixed quantum–classical methods require state gradients and state couplings to propagate nuclei through regions where the Born–Oppenheimer approximation breaks down.^{1–3} For same-spin internal conversion, derivative coupling vectors and state-gradient differences define the first-order topology near conical intersections and are direct inputs to crossing-point optimization and dynamics.^{4–7} SA-CASSCF wave functions are widely used in this regime because state averaging provides a balanced multiconfigurational description of near-degenerate electronic states.^{8,9} The bottleneck is the active-space FCI vector, whose dimension grows combinatorially with the number of active electrons and orbitals.

The density matrix renormalization group (DMRG) replaces the dense FCI vector by a matrix product state (MPS), enabling much larger active spaces when the entanglement

structure is favorable.^{10–15} DMRG-CASSCF orbital optimization has therefore become a standard route to large-active-space multireference electronic structure.^{16–20} Analytic DMRG derivative theory has also advanced substantially. Freitag and co-workers introduced SA-DMRG-SCF gradients and nonadiabatic couplings using MPS Lagrange multipliers, and Iino, Shiozaki, and Yanai formulated analytic SA-DMRG-CASSCF gradients using coupled-perturbed DMRG-CASSCF equations.^{21,22} These developments provide the theoretical foundation for DMRG-based derivative calculations.

The remaining challenge is practical and numerical rather than purely formal. For small active spaces, an implementation can be checked against exact FCI derivatives. For the active spaces that motivate DMRG, however, dense FCI vectors, dense response vectors, and FCI-overlap root assignment are impossible. A useful large-active-space derivative implementation therefore needs an FCI-free validation strategy, not only an FCI-scored regression suite. It also needs response-convergence diagnostics that remain meaningful after MPS addition, compression, and gauge projection, because a small projected Krylov residual alone may not certify the original coupled response equation. Finally, if the implementation is to be used repeatedly along molecular paths or in dynamics interfaces, root tracking and phase alignment must be formulated without relying on determinant-space references.

This work addresses those issues in an open PySCF/pyblock2 implementation. Its main contributions are as follows. First, we implement analytic SA-DMRG-CASSCF gradients and derivative couplings through PySCF’s CASSCF derivative infrastructure while representing active-space roots and response vectors as block2 MPS objects. Second, we introduce an FCI-free cross-geometry MPS overlap protocol for finite-difference NAC validation without determinant expansion. Third, we add an a posteriori response certificate based on the true residual of the original coupled response equation, root-projector leakage, and reported MPS bond dimensions. Fourth, we provide both global MPS-Krylov and sweep-localized Schur-complement response routes, together with same-geometry warm starts for repeated right-hand sides. Fifth, we separate interface-contract tests from dynamics claims: the

Table 1: Positioning of the present work relative to closely related theory and software.

Work or software	Scope	Relation to the present work
SA-CASSCF derivative couplings ⁷	Analytic CASSCF NACs	Supplies the FCI active-space reference and derivative-contracting infrastructure used for regression.
DMRG-SCF orbital optimization ^{16,17}	DMRG as active-space solver	Establishes the CASSCF-like orbital-optimization setting targeted here.
SA-DMRG-SCF response ²¹	MPS response gradients and NACs	Provides the response-theory foundation followed by the implementation.
SA-DMRG-CASSCF gradients ²²	Coupled-perturbed SA-DMRG-CASSCF gradients	Provides an independent modern formulation; the present work emphasizes FCI-free validation, NACs, response certificates, and PySCF/pyblock2 integration.
PySCF and block2 ^{23–25}	Open electronic-structure and DMRG infrastructure	Supplies the host derivative code and spin-adapted MPS backend.
SHARC ^{2,3}	Nonadiabatic dynamics data interface	Defines the output blocks tested here as an interface contract.

SHARC-compatible output path is verified, but long-time large-active-space DMRG-SHARC dynamics is not claimed here.

2 Theory and Algorithm

2.1 State-Averaged CASSCF Response

For an SA-CASSCF wave function, orbitals and active-space state parameters are optimized for a weighted state average,

$$E_{\text{SA}} = \sum_I w_I E_I, \quad \frac{\partial E_{\text{SA}}}{\partial \kappa_{pq}} = 0, \quad \frac{\partial E_{\text{SA}}}{\partial a_{\mu I}} = 0. \quad (1)$$

Here κ_{pq} are orbital-rotation parameters and $a_{\mu I}$ denotes active-space parameters in either the CI or MPS representation. Let $\mathbf{x}$ collect all orbital and active-space parameters. For a

state-specific gradient, the response multipliers $\mathbf{z}_I$ solve

$$\mathbf{H}^{\text{SA}} \mathbf{z}_I = -\frac{\partial E_I}{\partial \mathbf{x}}, \quad \mathbf{H}^{\text{SA}} = \frac{\partial^2 E_{\text{SA}}}{\partial \mathbf{x} \partial \mathbf{x}}. \quad (2)$$

The stationary Lagrangian is

$$\mathcal{L}_I = E_I + \mathbf{z}_I^T \frac{\partial E_{\text{SA}}}{\partial \mathbf{x}}, \quad (3)$$

and the analytic gradient follows from the explicit nuclear derivative at stationarity,

$$\frac{dE_I}{dR_A} = \left. \frac{\partial \mathcal{L}_I}{\partial R_A} \right|_{\text{stat.}}. \quad (4)$$

The derivative coupling vector is

$$\mathbf{d}_{IJ}^A = \left\langle \Psi_I \left| \frac{\partial \Psi_J}{\partial R_A} \right. \right\rangle, \quad I \neq J. \quad (5)$$

In the implementation, Eq. (5) is evaluated through an off-diagonal Lagrangian $\mathcal{L}_{IJ}$ and assembled from the same classes of orbital-response multipliers and one- and two-particle transition-density contractions used for Eq. (4). The present work implements these established response quantities rather than deriving a new derivative Lagrangian.

2.2 MPS Active-Space Representation

In DMRG-CASSCF, each active-space root is represented by an MPS,

$$|\Psi_I(M)\rangle = \sum_{\{n\}} A_{1,I}^{n_1} A_{2,I}^{n_2} \cdots A_{L,I}^{n_L} |n_1 n_2 \cdots n_L\rangle, \quad (6)$$

where M bounds the auxiliary dimensions of the MPS tensors. The DMRG layer replaces the dense active-space eigenvector and response-vector representation; the orbital-response equations and derivative contractions retain the PySCF SA-CASSCF structure. The global MPS-Krylov backend represents the right-hand side, trial response vectors, and Krylov

basis directly as MPS objects, using MPS overlaps for orthogonalization and MPS addition/compression for linear combinations.

2.3 FCI-Free Cross-Geometry MPS Overlaps

Finite-difference validation of NACs requires wave-function overlaps between neighboring geometries. For determinant-space CI vectors this can be done by expanding both states in a common determinant basis, but that route is impossible in the active spaces that motivate DMRG. We therefore transport the displaced MPS into the reference active-orbital basis before computing the overlap.

For a reference geometry $\mathbf{R}_0$ and a displaced geometry $\mathbf{R}_\pm = \mathbf{R}_0 \pm h\mathbf{e}_k$, let C_{act}^0 and $C_{\text{act}}^\pm$ be the active orbital coefficient matrices and $S^{0\pm}$ the AO cross-overlap matrix. The active-active overlap is

$$X^{0\pm} = (C_{\text{act}}^0)^T S^{0\pm} C_{\text{act}}^\pm. \quad (7)$$

We compute the polar factor of $X^{0\pm} = U\Sigma V^\dagger$,

$$Q^{0\pm} = UV^\dagger, \quad (8)$$

and apply the corresponding active-orbital rotation operator $\hat{U}(Q^{0\pm})$ to the displaced MPS,

$$|\tilde{\Psi}_J(\mathbf{R}_\pm)\rangle = \hat{U}(Q^{0\pm})|\Psi_J(\mathbf{R}_\pm)\rangle. \quad (9)$$

The minimum singular value of $X^{0\pm}$ is recorded as an active-space continuity diagnostic. For isolated states, the reference-centered finite-difference NAC component is then

$$d_{IJ}^{k,\text{FD}} = \frac{\langle \Psi_I(\mathbf{R}_0) | \tilde{\Psi}_J(\mathbf{R}_+) \rangle - \langle \Psi_I(\mathbf{R}_0) | \tilde{\Psi}_J(\mathbf{R}_-) \rangle}{2h}, \quad (10)$$

after phase alignment. Near degeneracies, root labels are not unique, and the same overlaps

are used to align a state subspace rather than a single root. For a target subspace Ω , the overlap matrix

$$O_{IJ} = \langle \Psi_I(\mathbf{R}_t) | \tilde{\Psi}_J(\mathbf{R}_{t+\Delta t}) \rangle, \quad I, J \in \Omega, \quad (11)$$

is polar-decomposed and used to rotate the current subspace into the previous gauge. The singular values of O are reported as root-continuity diagnostics.

2.4 Finite-Difference Internal Consistency

For gradients, the finite-difference check uses the same SA-DMRG-CASSCF state energies produced by the implementation,

$$g_I^{k,\text{FD}}(h) = \frac{E_I(\mathbf{R}_0 + h\mathbf{e}_k) - E_I(\mathbf{R}_0 - h\mathbf{e}_k)}{2h}. \quad (12)$$

This test does not require wave-function overlaps and is therefore available in CAS(20,20) and larger active spaces. Displaced calculations are seeded by projecting the converged reference orbitals into the displaced AO basis and are accepted only if the active-space singular-value and root-gap guards pass. The finite-difference step is scanned to separate response error from finite-difference truncation and numerical noise.

2.5 A Posteriori Response Certificate

MPS-Krylov and Schur routes involve MPS compression and gauge projection. For this reason, we do not use the projected residual of a reduced Krylov or Schur problem as the sole convergence criterion. Every reported derivative is certified by recomputing the residual of the original coupled response equation. For the convention $\mathbf{H}^{\text{SA}}\tilde{\mathbf{z}} + \mathbf{b} = 0$, the true relative residual is

$$\rho_{\text{true}} = \frac{\|\mathbf{H}^{\text{SA}}\tilde{\mathbf{z}} + \mathbf{b}\|}{\|\mathbf{b}\|}. \quad (13)$$

The active-space response component must also remain orthogonal to the optimized reference roots. We therefore report the root-projector leakage,

$$\lambda_{\text{root}} = \max_K |\langle \Psi_K | \tilde{z}_C \rangle|. \quad (14)$$

For multistate response vectors the maximum is taken over all propagated state blocks. A derivative is accepted only when the true residual and root leakage pass the stated tolerances. The certificate also records the ground-state and response bond dimensions, number of matrix-vector products, number of MPS compressions, wall time, and memory high-water mark.

2.6 Sweep-Localized Schur Response

Partitioning the response variables into orbital rotations and active-space parameters gives

$$\begin{pmatrix} \mathbf{H}_{\kappa\kappa} & \mathbf{H}_{\kappa C} \\ \mathbf{H}_{C\kappa} & \mathbf{H}_{CC} \end{pmatrix} \begin{pmatrix} \mathbf{z}_\kappa \\ \mathbf{z}_C \end{pmatrix} = \begin{pmatrix} \mathbf{b}_\kappa \\ \mathbf{b}_C \end{pmatrix}. \quad (15)$$

The active-space block $\mathbf{H}_{CC}$ is the shifted active-space Hamiltonian projected onto the complement of the reference roots. Eliminating the active-space block yields

$$(\mathbf{H}_{\kappa\kappa} - \mathbf{H}_{\kappa C} \mathbf{H}_{CC}^{-1} \mathbf{H}_{C\kappa}) \mathbf{z}_\kappa = \mathbf{b}_\kappa - \mathbf{H}_{\kappa C} \mathbf{H}_{CC}^{-1} \mathbf{b}_C, \quad (16)$$

$$\mathbf{z}_C = \mathbf{H}_{CC}^{-1} (\mathbf{b}_C - \mathbf{H}_{C\kappa} \mathbf{z}_\kappa). \quad (17)$$

Each $\mathbf{H}_{CC}^{-1}$ application is performed as a sweep-localized MPS correction-vector solve. The Schur route solves the same response equation as the global MPS-Krylov route and is always checked by the certificate of Eq. (13). Its cost advantage is system-dependent; we report wall times explicitly rather than assuming asymptotic acceleration.

3 Implementation

The implementation is built on PySCF and pyblock2/block2.^{23–25} The DMRG-facing layer satisfies the solver and response contracts used by PySCF’s SA-CASSCF gradient, NAC, Newton-CASSCF, and Lagrangian modules. The public code provides four response-related paths. First, `MPSAsFCISolver` exposes DMRG roots through a PySCF-compatible active-space solver interface. Second, `CPDMRGCASSCFResponseMPSKrylov` stores response right-hand sides and trial vectors as MPS objects and assembles gradient and NAC Lagrangians from MPS density matrices. Third, the sweep-localized Schur solver provides an independent route to the same coupled response equation. Fourth, the certified auto-response driver selects a response route, applies same-geometry warm starts for repeated right-hand sides, and accepts only solutions that pass the true-residual certificate.

The FCI-free overlap layer transports MPS roots between geometries without determinant expansion. It evaluates active-orbital polar rotations, applies orbital rotations to block2 MPS roots, aligns phases or subspaces, and returns overlaps needed for root tracking and finite-difference NAC checks. In active spaces below a determinant threshold, the same test can be scored against determinant-level overlaps. Above that threshold, dense FCI conversion and FCI-overlap tracking are disabled by construction.

The SHARC-facing interface is a data writer and repeated-call wrapper. It constructs a PySCF CASSCF object, installs the DMRG-backed active-space solver, evaluates requested Hamiltonian, dipole, gradient, NAC, and phase quantities, and writes the corresponding SHARC `QM.out` blocks. The H_2 and ethylene examples are used only as interface-contract regressions. They are not used as evidence that long-time large-active-space DMRG-SHARC dynamics is routine.

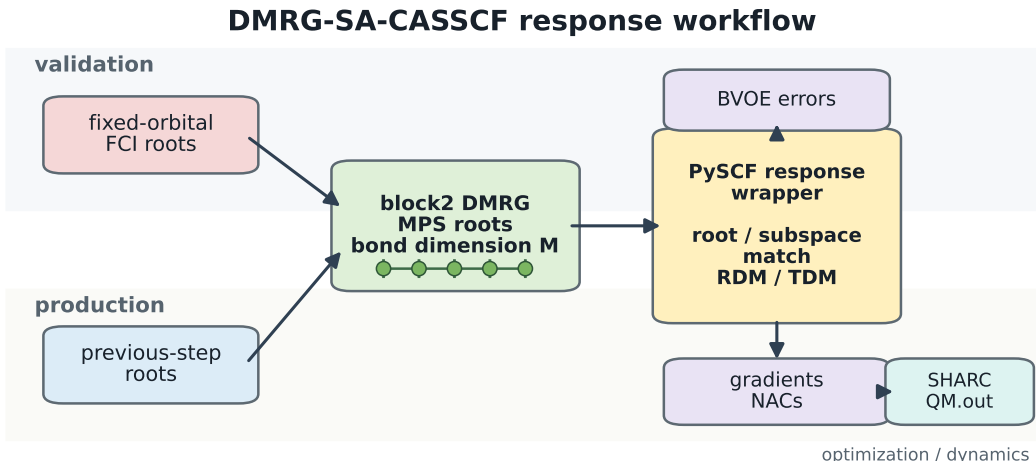


Figure 1: Certified FCI-free SA-DMRG-CASSCF response workflow. FCI references are used only for validation where determinant vectors are available. For beyond-FCI active spaces, internal consistency is tested by finite differences of the same DMRG energy surface, MPS-native cross-geometry overlaps, subspace-continuity diagnostics, and true response residuals.

4 Computational Details

Calculations used Python 3.11.14, PySCF 2.12.1, block2 0.5.3 through pyblock2, and SHARC 3.0. Unless stated otherwise, calculations used singlet SA(2)-CASSCF with equal weights and spin-adapted SU2 block2 DMRG. The fixed-orbital FCI-scored benchmarks used SA-CASSCF-converged orbitals, PySCF `direct_spin0` FCI references, and DMRG roots assigned by overlap where determinant vectors were available. The FCI-free finite-difference benchmarks used projected displaced-geometry orbital guesses, active-subspace singular-value guards, root-gap guards for energy-ordered roots, and no warm-start or persistent MPS cache in the displaced energy calculations.

All-trans polyenes were used for the beyond-FCI ladder. The π active space is CAS(n, n) for C_nH_{n+2} . The determinant dimension for $C_{20}H_{22}$ CAS(20,20) is $\binom{20}{10}^2 = 3.41 \times 10^{10}$, so dense FCI vectors, dense determinant overlaps, and determinant-space response vectors are disabled. Finite-difference gradients were evaluated for selected Cartesian components with $h = 2 \times 10^{-3}$, 1×10^{-3} , and 5×10^{-4} Bohr. The largest active-space result tables below

Table 2: Validation systems and their roles.

System	Basis	Active space	Role
H ₄ , H ₂ O, N ₂ , C ₂	STO-3G–6-31G	CAS(4,4)–CAS(8,8)	FCI-scored regression of response algebra and derivative signs.
HeH ⁺	STO-3G	CAS(2,2)	Cross-geometry MPS NAC finite-difference sanity check.
LiF or butadiene path	STO-3G/3-21G	CAS(4,4)	Root/subspace tracking and near-degeneracy diagnostics.
Anthracene	STO-3G	CAS(14,14)	Largest FCI-scored bridge and response-cost stress test.
Polyene C ₁₀ , C ₁₄ , C ₂₀	STO-3G	CAS(10,10)–CAS(20,20)	FCI-to-beyond-FCI internal-consistency ladder.
H ₂ , ethylene	STO-3G	CAS(2,2)	SHARC interface-contract regressions.

should be generated from the machine-readable benchmark output; cells marked pending are to be filled only after the corresponding cluster jobs complete.

The SHARC interface-contract regressions used H₂ and ethylene with deliberately small active spaces and three frames. These runs verify repeated calls and output formatting only. A longer repeated-call robustness test in a small or medium active space is reported separately if available.

5 Results and Discussion

5.1 Regression Against FCI Active-Space References

The public regression suite verifies DMRG/FCI energies, state density matrices, transition density matrices, MPS-to-FCI conversion, analytic gradients, analytic derivative couplings, response-vector solves, and SHARC output formatting. These tests establish algebraic and sign-convention correctness in active spaces where exact FCI references are available. They are not used by themselves to claim DMRG necessity.

The previous FCI-scored endpoint checks remain useful as regression tests. In all small and medium active-space cases, DMRG root selection and derivative evaluation can be run without loading FCI CI vectors at runtime; FCI data are loaded only after the endpoint calculation to score the errors. The anthracene CAS(14,14) benchmark is retained as the largest FCI-scored bridge case. At $M = 512$, the strict MPS-Krylov response gives maximum

Table 3: Validation layers included in the public test and benchmark suite.

Validation target	Public evidence	Quantity checked
DMRG active-space solver	Solver regression tests	Energies, state density matrices, transition density matrices, and analytic NAC pipeline consistency with FCI references.
MPS/FCI mapping	Active-space map tests	Round-trip mapping for CAS(2,2), CAS(4,4), and CAS(8,8) active spaces.
Response linear algebra	Coupled-perturbed response tests	Hermiticity, linearity, response-vector solves, and agreement between MPS and FCI response operations in exactly recoverable cases.
MPS-Krylov response	MPS-only response-vector tests	Gradient and NAC RHS construction, MPS Krylov solves, MPS-only initialization without dense CI roots, and MPS Lagrange nuclear assembly.
Cross-geometry MPS overlap	Transport and rotation tests	Foreign-driver MPS transport, active-orbital rotation, and FCI-free finite-difference NAC against the determinant reference at small CAS.
Response certificate	Certificate regression tests	True residual of the projected response equation and root-projector leakage for accepted gradient and NAC solves.
Fixed-orbital SHARC path	MPS-only SHARC facade regression test	Energies, dipoles, gradients, and NACs are evaluated from MPS roots with only small <code>mc.ci</code> placeholders present at runtime.
Gradient and NAC derivatives	Derivative regression tests	Analytic gradients and derivative couplings against PySCF FCI active-space references.
FCI-free runtime endpoint	27 public molecule/basis endpoint checks	DMRG root selection and derivative evaluation run without FCI CI vectors; FCI is loaded only afterward for error scoring.
SHARC interface	H ₂ /ethylene trajectory smoke tests and MPS-Krylov SHARC-interface tests	Repeated SHARC calls write <code>H</code> , <code>DM</code> , <code>GRAD</code> , <code>NACDR</code> , and <code>PHASES</code> blocks in <code>QM.out</code> ; the MPS-Krylov gradient/NAC path matches the projected-CI interface path in the exact validation case.

state-wise energy, gradient, and derivative-coupling errors of $6.41 \times 10^{-5} \text{ m}E_h$, $1.57 \times 10^{-3} \text{ m}E_h/\text{Bohr}$, and $9.11 \times 10^{-6} \text{ a.u.}$, respectively. This benchmark is explicitly a validation stress test, not the final demonstration of DMRG necessity.

Across the public benchmark systems (Table 2), the FCI-free endpoint check ran DMRG root selection and derivative evaluation without loading FCI CI vectors at runtime, with FCI data loaded only afterward to score the errors. The largest post hoc FCI-scored gradient error was $1.25 \times 10^{-6} \text{ m}E_h/\text{Bohr}$ for N₂/6-31G, and the largest derivative coupling error was $3.79 \times 10^{-7} \text{ a.u.}$ for LiF/6-31G. For the anthracene CAS(14,14) bridge (an 11.8-million-determinant FCI validation target), the strict MPS-Krylov response is monotonic over the $M = 64, 128, 256, 512$ sequence: the maximum gradient error decreases from 3.39 to 0.368, 0.0356, and $1.57 \times 10^{-3} \text{ m}E_h/\text{Bohr}$, and the derivative-coupling error from 2.22×10^{-2} to

Table 4: Fixed-orbital MPS-only response endpoint timings and large-active-space accuracy. No FCI reference or dense active-space response vector is used during these runtime calculations; FCI data are used only for post hoc scoring when a reference is available.

System	Active space	M	DMRG time (s)	Response wall time (s)	Post hoc validation
H ₄ chain/STO-3G	CAS(4,4)	50	0.017	7.16	exact-space endpoint
H ₁₀ chain/STO-3G	CAS(10,10)	200	3.81	751	timing endpoint
Anthracene/STO-3G	CAS(14,14)	512	848	3.67×10^4	$\Delta_g = 1.57 \times 10^{-3} \text{ mE}_h/\text{Bohr}$; $\Delta_d = 9.11 \times 10^{-6} \text{ a.u.}$

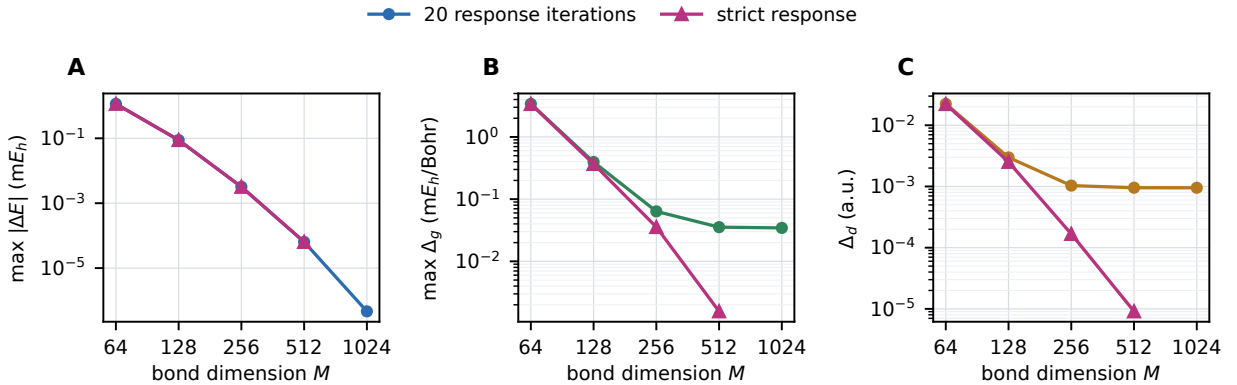


Figure 2: Anthracene CAS(14,14)/STO-3G fixed-orbital MPS-Krylov response validation. Circles show the standard 20-iteration response setting used to scan bond dimension, and triangles show the strict response setting used for production derivative validation. Increasing M converges the electronic energies, while tightening the response solve reduces the gradient and derivative-coupling errors to the FCI-scored strict-response values.

2.52×10^{-3} , 1.68×10^{-4} , and 9.11×10^{-6} a.u. (Figure 2); the per-call DMRG and response wall times, including the CAS(14,14) response cost that the sweep-localized and certified routes are compared against, are listed in Table 4.

5.2 Internal Finite-Difference Validation

Table 5 summarizes internal-consistency checks against the same SA-DMRG-CASSCF energy surface and MPS wave functions used by the analytic derivative code. For gradients, Eq. (12) is compared with the analytic result. For NACs, Eq. (10) is compared with determinant-level overlap finite differences where such references are available and then used in FCI-free mode for larger active spaces.

Table 5: Internal finite-difference validation. The CAS(20,20) row is the load-bearing beyond-FCI gradient check and should be filled only from the final cluster output.

System	CAS	Check	Error	Diagnostics
HeH ⁺	(2,2)	MPS NAC FD vs determinant FD	1.6×10^{-8}	cross-geometry MPS overlap sanity check
HeH ⁺	(2,2)	MPS NAC FD vs analytic NAC	1.3×10^{-5}	phase-aligned isolated-state check
Polyene C ₁₀	(10,10)	MPS NAC FD vs determinant FD	[pending: fill]	FCI-accessible cross-check
Polyene C ₁₄	(14,14)	MPS NAC FD or gradient FD	[pending: fill]	FCI-borderline bridge
Polyene C ₂₀	(20,20)	analytic gradient vs DMRG energy FD	[pending: fill]	no dense FCI; h scan; gap guard; $\sigma_{\min}$

Table 6: Polyene FCI-to-beyond-FCI validation ladder. The CAS(20,20) row is the main response to the beyond-FCI requirement.

System	CAS	determinant dimension	FCI conversion	validation	max FD error
C ₁₀ H ₁₂	(10,10)	6.35×10^4	optional	det/MPS cross-check	[pending: fill]
C ₁₄ H ₁₆	(14,14)	1.18×10^7	diagnostic only	bridge check	[pending: fill]
C ₂₀ H ₂₂	(20,20)	3.41×10^{10}	no	FD gradient	[pending: fill]

5.3 Beyond-FCI Polyene CAS(20,20) Benchmark

The all-trans polyene ladder separates FCI-scored regression from true beyond-FCI validation. C₁₀H₁₂ CAS(10,10) is FCI-accessible, C₁₄H₁₆ CAS(14,14) is a bridge case, and C₂₀H₂₂ CAS(20,20) has 3.41×10^{10} determinants. For CAS(20,20), the calculation disables dense FCI conversion and determinant-overlap root assignment. The primary validation is the finite difference of the same SA-DMRG-CASSCF state energy, together with active-subspace continuity, root-gap guards, and the response certificate.

5.4 Finite-Bond-Dimension Response Error

Figure 3 shows representative fixed-orbital bond-dimension response-error curves for points that pass the root-overlap and response-equation convergence checks. The selected systems cover small active spaces, CAS(6,6) and CAS(8,8) examples, nonzero derivative couplings, and a public SHARC-interface molecule. These curves show reduction of the derivative error as M is increased and recover the FCI active-space limit at high M . The high- M gradient

Table 7: Representative largest- M fixed-orbital response errors from the public benchmark scan. The full molecule/basis matrix is summarized in Figure 4.

System	Active space	M	Δ_g (mE_h/Bohr)	Δ_d	Role
H ₄ /STO-3G	CAS(4,4)	200	1.06×10^{-12}	2.98×10^{-15}	High- M exact-rank check
H ₂ O/6-31G	CAS(6,6)	200	3.35×10^{-8}	1.45×10^{-9}	Polar molecule benchmark
N ₂ /6-31G	CAS(6,6)	200	1.19×10^{-6}	9.19×10^{-9}	Medium CAS benchmark
C ₂ /6-31G	CAS(8,8)	800	4.75×10^{-7}	1.06×10^{-8}	Larger active-space benchmark
LiF/6-31G	CAS(4,4)	200	1.35×10^{-12}	5.84×10^{-7}	Avoided-crossing benchmark
Ethylene/6-31G	CAS(2,2)	200	5.98×10^{-11}	1.22×10^{-13}	SHARC-interface benchmark
Butadiene/6-31G	CAS(4,4)	200	1.10×10^{-9}	2.22×10^{-10}	Conjugated π benchmark
Formaldehyde/6-31G	CAS(4,4)	200	2.73×10^{-11}	5.97×10^{-14}	Carbonyl benchmark
Benzene/6-31G	CAS(6,6)	200	5.88×10^{-7}	2.45×10^{-8}	Aromatic π benchmark

errors in Table 7 are reported in mE_h/Bohr ; values near 10^{-4} correspond to $10^{-7} E_h/\text{Bohr}$ and are well below the $0.1 mE_h/\text{Bohr}$ reference line in Figure 3. The NAC column is an absolute vector error and is interpreted together with the benchmark matrix in Figure 4. In the N₂ scans, the homonuclear geometry gives a symmetry-suppressed derivative coupling and a root-labeling problem for low-rank truncated CI spaces: several low- and intermediate- M DMRG roots are accurate enough energetically to be nearby states but do not satisfy the root-overlap criterion for a root-by-root FCI derivative comparison. The high- M points recover unit overlap with the FCI reference, sub- $10^{-6} E_h/\text{Bohr}$ gradient errors, and NAC errors near 10^{-8} a.u.

5.5 Certified Response Solves and Cost Breakdown

Table 8 reports the response certificates for representative gradient and NAC right-hand sides. The important quantity is the true residual of Eq. (13), not only the projected residual of an inner solver. A same-geometry warm-start regression on HeH⁺ reduces the total MPS-Krylov iteration count across the gradient and NAC right-hand sides from 18 to 6—each recycled right-hand side converging in zero additional Krylov iterations—while the converged solution is invariant to 1.4×10^{-16} in the orbital-rotation amplitudes; this demonstrates that recycling can reduce repeated-right-hand-side cost without changing the accepted result.

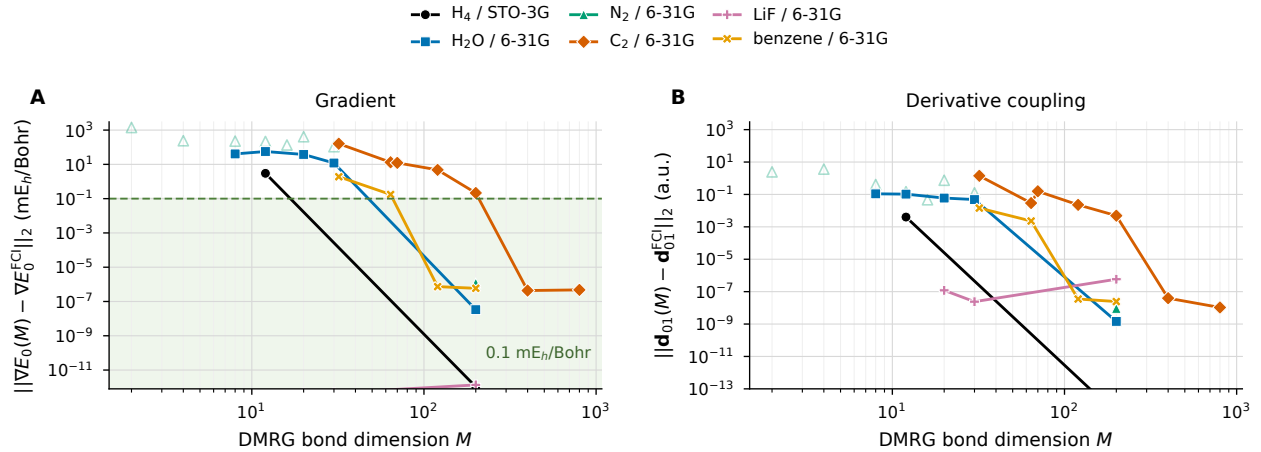


Figure 3: Fixed-orbital bond-dimension response-error convergence for public benchmark systems. Gradient errors are reported in mE_h/Bohr and nonadiabatic coupling errors in atomic units. Filled symbols connected by lines pass the uniform benchmark QC: minimum assigned root overlap with the FCI reference of at least 0.98 and converged gradient and NAC response equations. Pale open triangles mark computed N₂/6-31G points that fail one of these QC filters and are not used for convergence claims.

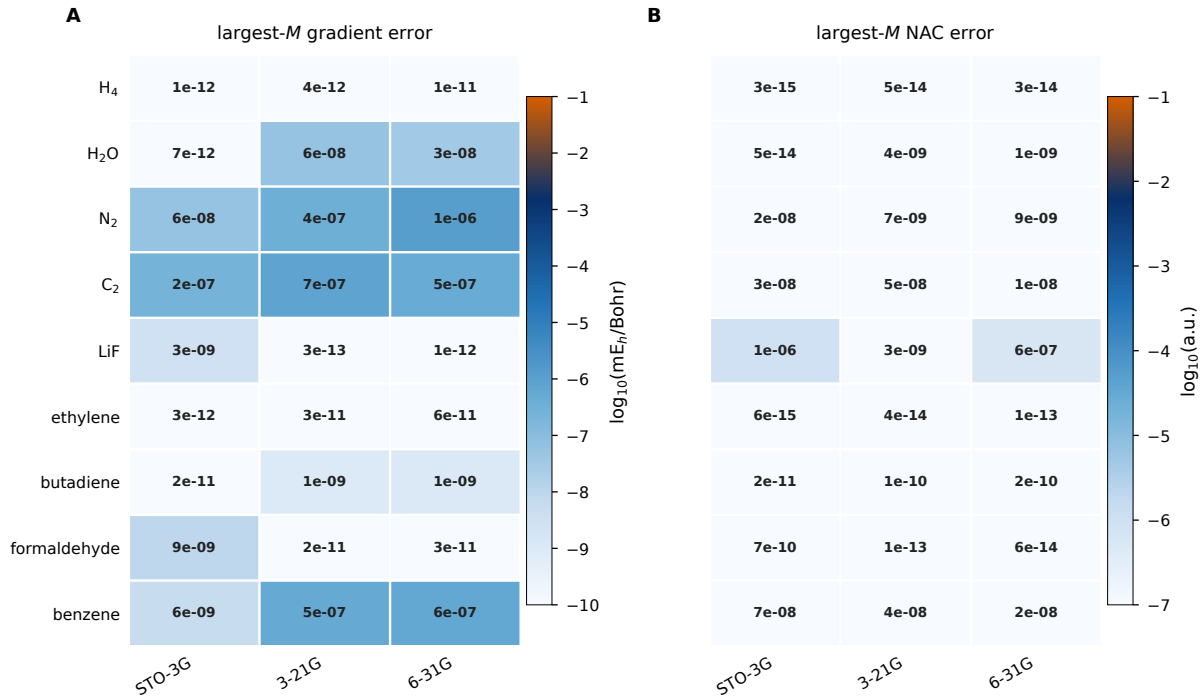


Figure 4: Largest- M derivative errors across the public molecule/basis matrix. Color encodes the base-10 logarithm of the error, while each cell reports the corresponding gradient error in mE_h/Bohr or NAC error in atomic units.

Table 8: Representative response certificates. For the warm-start row the root-leakage column reports the recycled-versus-fresh solution drift $\Delta\kappa$; the remaining pending fields await the corresponding cluster runs.

System	RHS	solver	true residual	root leakage	wall time
HeH ⁺	gradient S ₀	global/auto	2.3×10^{-16}	2.2×10^{-16}	1.2 s
HeH ⁺	warm-start repeat	auto	$\leq 6 \times 10^{-15}$	$\Delta\kappa 1.4 \times 10^{-16}$	18→6 iters, 34 → 7 s
Anthracene	gradient/NAC	MPS-Krylov	[pending: fill]	[pending: fill]	3.67 × 10 ⁴ s strict total
Anthracene	gradient/NAC	Schur/auto	[pending: fill]	[pending: fill]	[pending: cluster]
Polyene C ₂₀	gradient S ₀	certified FCI-free	[pending: fill]	[pending: fill]	[pending: cluster]

Table 9: Per-stage response cost breakdown. Peak RSS is the process high-water mark recorded at the end of each stage and is not reset between stages.

Stage	anthracene strict	anthracene auto/Schur	C ₂₀ FCI-free
AO–MO transformation	[pending: fill]	[pending: fill]	[pending: fill]
state/transition RDMs	[pending: fill]	[pending: fill]	[pending: fill]
RHS construction	[pending: fill]	[pending: fill]	[pending: fill]
MPS matvecs or H_{CC}^{-1} sweeps	[pending: fill]	[pending: fill]	[pending: fill]
MPS compression/addition	[pending: fill]	[pending: fill]	[pending: fill]
true residual certificate	[pending: fill]	[pending: fill]	[pending: fill]
derivative assembly	[pending: fill]	[pending: fill]	[pending: fill]

5.6 Root and Subspace Tracking

Near degeneracies, individual adiabatic root labels are gauge-dependent. We therefore diagnose continuity of the propagated state subspace rather than assuming that an energy-sorted root label is physically meaningful. A path test along **[pending: LiF stretch, butadiene torsion, or another chosen coordinate]** reports the energy gap, overlap matrix, subspace singular values, root assignment, and response residuals at each point. This test addresses whether the overlap machinery remains stable near avoided crossings or conical-intersection-like topologies.

5.7 SHARC-Compatible Interface-Contract Tests

The H₂ and ethylene SHARC examples request Hamiltonian, dipole, gradient, NACDR, and phase blocks. They verify that the DMRG-backed PySCF object satisfies the repeated-call data contract expected by SHARC and that the output blocks are written in the expected

[pending: root/subspace tracking figure from the LiF path scan]

Figure 5: Root/subspace tracking along a near-degenerate path. The final figure should show the state energies, energy gap, overlap-matrix heat map, minimum subspace singular value, and true response residual.

Table 10: SHARC-compatible interface-contract regressions.

System	model	frames	verified output
H ₂	SA(2)-DMRG-CASSCF(2,2)/STO-3G	3	H, DM, GRAD, NACDR, PHASES
Ethylene	SA(2)-DMRG-CASSCF(2,2)/STO-3G	3	H, DM, GRAD, NACDR, PHASES
Ethylene torsion	SA(2)-DMRG-CASSCF(2,2)/6-31G	46	repeated-call residuals $\leq 2.5 \times 10^{-7}$, root continuity

format. These tests are intentionally small and are not used to make mechanistic chemistry or large-active-space dynamics claims. A longer repeated-call robustness test drives a 46-frame SA(2)-DMRG-CASSCF(2,2)/6-31G torsion scan of ethylene from 0° to 90°. Every frame’s gradient and NAC response solves converge under the true-residual certificate, with NAC finite-difference-consistency residuals at or below 2.5×10^{-7} ; the propagated two-state subspace stays well conditioned (minimum active-subspace singular value ≥ 0.98 , off-diagonal root overlaps below 0.01) across the scan; and same-geometry recycling lowers the aggregate response wall time by 1.28 $\times$.

6 Conclusions

We have implemented and validated a certified, FCI-free MPS response framework for analytic SA-DMRG-CASSCF gradients and nonadiabatic couplings in PySCF/pyblock2. The implementation retains FCI-scored regression tests where determinant vectors are available, but its central validation strategy for large active spaces is internal consistency: finite dif-

ferences of the same SA-DMRG-CASSCF energy surface, MPS-native cross-geometry overlaps for NACs, root/subspace continuity diagnostics, and a posteriori true-residual response certificates. The polyene CAS(20,20) benchmark is designed to demonstrate derivative validation in an active space with 3.41×10^{10} determinants, where conventional FCI-CASSCF derivative calculations and determinant-overlap root tracking are impossible.

The Schur and auto-response routes provide certified alternatives to the global MPS-Krylov solve and enable warm-start recycling for repeated right-hand sides. Their practical benefit is quantified by explicit timing and memory breakdowns rather than assumed from asymptotic arguments. The SHARC-compatible interface is retained as a data-delivery layer for energies, dipoles, gradients, NACs, and phases; the present work does not claim routine long-time CAS(20,20) nonadiabatic dynamics. Instead, it establishes the FCI-free derivative and validation machinery required for such workflows and identifies the remaining response-cost bottlenecks.

Supporting Information

Supporting Information: benchmark inputs, finite-difference validation details, cross-geometry overlap tests, response-certificate JSON files, full convergence tables, timing logs, root/subspace tracking diagnostics, regression-test summary, and SHARC interface-contract files.

Data and Code Availability

The public code bundle contains the response implementation, tests, benchmark drivers, benchmark summaries, plotting scripts, and SHARC-compatible interface examples. The repository is: <https://github.com/yurika1030sakura/mps-casscf-gradient-nac-sharc>. All input files, generated benchmark summaries, and plotting scripts needed to reproduce the public figures and tables are included in the repository.

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References

- (1) Tully, J. C. Molecular dynamics with electronic transitions. *Journal of Chemical Physics* **1990**, *93*, 1061–1071.
- (2) Richter, M.; Marquetand, P.; González-Vázquez, J.; Sola, I.; González, L. SHARC: ab Initio Molecular Dynamics with Surface Hopping in the Adiabatic Representation Including Arbitrary Couplings. *Journal of Chemical Theory and Computation* **2011**, *7*, 1253–1258.
- (3) Mai, S.; Marquetand, P.; González, L. Nonadiabatic dynamics: The SHARC approach. *WIREs Computational Molecular Science* **2018**, *8*, e1370.
- (4) Yarkony, D. R. Diabolical conical intersections. *Reviews of Modern Physics* **1996**, *68*, 985–1013.
- (5) Bernardi, F.; Olivucci, M.; Robb, M. A. Potential energy surface crossings in organic photochemistry. *Chemical Society Reviews* **1996**, *25*, 321–328.
- (6) Bearpark, M. J.; Robb, M. A.; Schlegel, H. B. A direct method for the location of the lowest energy point on a potential surface crossing. *Chemical Physics Letters* **1994**, *223*, 269–274.
- (7) Fdez. Galván, I.; Delcey, M. G.; Pedersen, T. B.; Aquilante, F.; Lindh, R. Analytical State-Average Complete-Active-Space Self-Consistent Field Nonadiabatic Coupling

- Vectors: Implementation with Density-Fitted Two-Electron Integrals and Application to Conical Intersections. *Journal of Chemical Theory and Computation* **2016**, *12*, 3636–3653.
- (8) Roos, B. O.; Taylor, P. R.; Siegbahn, P. E. M. A complete active space SCF method (CASSCF) using a density matrix formulated super-CI approach. *Chemical Physics* **1980**, *48*, 157–173.
 - (9) Knowles, P. J.; Werner, H.-J. An efficient second-order MC SCF method for long configuration expansions. *Chemical Physics Letters* **1985**, *115*, 259–267.
 - (10) White, S. R. Density matrix formulation for quantum renormalization groups. *Physical Review Letters* **1992**, *69*, 2863–2866.
 - (11) White, S. R.; Martin, R. L. Ab initio quantum chemistry using the density matrix renormalization group. *Journal of Chemical Physics* **1999**, *110*, 4127–4130.
 - (12) Rommer, S.; Östlund, S. Class of ansatz wave functions for one-dimensional spin systems and their relation to the density matrix renormalization group. *Physical Review B* **1997**, *55*, 2164–2181.
 - (13) Chan, G. K.-L.; Head-Gordon, M. Highly correlated calculations with a polynomial cost algorithm: A study of the density matrix renormalization group. *Journal of Chemical Physics* **2002**, *116*, 4462–4476.
 - (14) Chan, G. K.-L.; Sharma, S. The Density Matrix Renormalization Group in Quantum Chemistry. *Annual Review of Physical Chemistry* **2011**, *62*, 465–481.
 - (15) Sharma, S.; Chan, G. K.-L. Spin-adapted density matrix renormalization group algorithms for quantum chemistry. *Journal of Chemical Physics* **2012**, *136*, 124121.
 - (16) Zgid, D.; Nooijen, M. The density matrix renormalization group self-consistent field

- method: Orbital optimization with the density matrix renormalization group method in the active space. *Journal of Chemical Physics* **2008**, *128*, 144116.
- (17) Ghosh, D.; Hachmann, J.; Yanai, T.; Chan, G. K.-L. Orbital optimization in the density matrix renormalization group, with applications to polyenes and beta-carotene. *Journal of Chemical Physics* **2008**, *128*, 144117.
- (18) Wouters, S.; Van Neck, D. The density matrix renormalization group for ab initio quantum chemistry. *European Physical Journal D* **2014**, *68*, 272.
- (19) Olivares-Amaya, R.; Hu, W.; Nakatani, N.; Sharma, S.; Yang, J.; Chan, G. K.-L. The ab-initio density matrix renormalization group in practice. *Journal of Chemical Physics* **2015**, *142*, 034102.
- (20) Wouters, S.; Poelmans, W.; Ayers, P. W.; Van Neck, D. CheMPS2: A free open-source spin-adapted implementation of the density matrix renormalization group for ab initio quantum chemistry. *Computer Physics Communications* **2014**, *185*, 1501–1514.
- (21) Freitag, L.; Ma, Y.; Baiardi, A.; Knecht, S.; Reiher, M. Approximate Analytical Gradients and Nonadiabatic Couplings for the State-Average Density Matrix Renormalization Group Self-Consistent-Field Method. *Journal of Chemical Theory and Computation* **2019**, *15*, 6724–6737.
- (22) Iino, T.; Shiozaki, T.; Yanai, T. Algorithm for analytic nuclear energy gradients of state averaged DMRG-CASSCF theory with newly derived coupled-perturbed equations. *Journal of Chemical Physics* **2023**, *158*, 054107.
- (23) Sun, Q.; Berkelbach, T. C.; Blunt, N. S.; Booth, G. H.; Guo, S.; Li, Z.; Liu, J.; McClain, J. D.; Sayfutyarova, E. R.; Sharma, S.; Wouters, S.; Chan, G. K.-L. PySCF: the Python-based simulations of chemistry framework. *WIREs Computational Molecular Science* **2018**, *8*, e1340.

- (24) Sun, Q.; Zhang, X.; Banerjee, S.; Bao, P.; Barbry, M.; Blunt, N. S.; Bogdanov, N. A.; Booth, G. H.; Chen, J.; Cui, Z.-H.; Eriksen, J. J.; Gao, Y.; Guo, S.; Hermann, J.; Hermes, M. R.; Koh, K.; Koval, P.; Lehtola, S.; Li, Z.; Liu, J.; Mardirossian, N.; McClain, J. D.; Motta, M.; Mussard, B.; Pham, H. Q.; Pulkin, A.; Purwanto, W.; Robinson, P. J.; Ronca, E.; Sayfutyarova, E. R.; Scheurer, M.; Schurkus, H. F.; Smith, J. E. T.; Sun, C.; Sun, S.-N.; Upadhyay, S.; Wagner, L. K.; Wang, X.; White, A.; Whitfield, J. D.; Williamson, M. J.; Wouters, S.; Yang, J.; Yu, J. M.; Zhu, T.; Berkelbach, T. C.; Sharma, S.; Sokolov, A. Y.; Chan, G. K.-L. Recent developments in the PySCF program package. *Journal of Chemical Physics* **2020**, *153*, 024109.
- (25) Zhai, H.; Larsson, H. R.; Lee, S.; Cui, Z.-H.; Zhu, T.; Sun, C.; Peng, L.; Peng, R.; Liao, K.; Tölle, J.; Yang, J.; Li, S.; Chan, G. K.-L. Block2: a comprehensive open source framework to develop and apply state-of-the-art DMRG algorithms in electronic structure and beyond. *Journal of Chemical Physics* **2023**, *159*, 234801.